

Saturated Superfluid Vacuum III

Irreversible Time and Wake Entropy

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Abstract

This paper addresses only the origin of irreversible time in the Saturated Superfluid Vacuum (SSV) framework. We argue that time is not a background dimension through which reality moves, but the local rate of change of the vacuum order parameter Ψ . The past does not exist as a destination or region; it survives only as wake structure encoded in the present state of the medium. Events in Ψ do not simply occur and vanish. They deposit distributed traces in the form of phonon phases, vortex distortions, Kelvin-wave content, density correlations, and altered future reconnection opportunities. We therefore propose that entropy in SSV should not be defined as a bare event count, but as the persistent wake-complexity of the present state.

On this view, irreversible time is not the flow of an external parameter, but the one-way accumulation and redistribution of wake structure in Ψ . Formal time reversal in equations is distinguished from physical reversal in the medium: the substitution $t \rightarrow -t$ defines a mathematical transformation on trajectories, but does not supply a physical operation that globally unwrites the distributed wake of an event. This dissolves the usual paradoxes of time travel by treating them as category errors arising from the reification of the past as a place. The paper is deliberately narrow in scope. It does not attempt black-hole mechanics, cosmology, or the cosmic microwave background; those belong in later papers.

1 Introduction

The standard treatment of time in fundamental physics is powerful, but conceptually unstable. At the level of formal equations, time often appears as a reversible parameter. At the level of experience and thermodynamics, it appears as an irreversible order of change. This tension is usually managed rather than resolved.

The aim of this paper is narrow. We do not attempt here to derive gravity, black holes, cosmology, or galactic structure. We address only the origin of irreversible time in the SSV framework. The central claim is:

Time is the local rate of change of Ψ , the past survives only as wake structure in the present state, and entropy is the persistent wake-complexity of that state.

This paper is therefore about ontology first and thermodynamic interpretation second. The goal is to state the time claim cleanly enough that the later papers can inherit it without having to carry the whole argument again.

1.1 What This Paper Does Not Attempt

Because the SSV program is broad, it is important to state the boundary conditions of the present paper explicitly. This paper does not attempt:

- a black-hole ontology,

- a derivation of gravitational dynamics,
- a cosmological account of the CMB,
- or a general survey of all testable predictions in the framework.

Those topics matter, but carrying them here would blur the argument. The task of the present paper is to make the time claim stand on its own.

1.2 Roadmap

Section 2 states the minimal SSV setting used in this paper. Section 3 states the ontological claim that time is local change rather than a background dimension. Section 4 develops the idea that events write wake structure into the medium. Section 5 reframes entropy as wake-complexity. Section 6 distinguishes formal time reversal in equations from physical reversal in the medium. Section 7 treats time-travel paradoxes as category errors. Section 8 closes with the remaining open technical questions.

2 Minimal SSV Setting

Only a small part of the SSV framework is needed here. The physical vacuum is represented by a macroscopic order parameter

$$\Psi(\mathbf{x}, t) = \sqrt{\rho(\mathbf{x}, t)} e^{i\theta(\mathbf{x}, t)}, \quad (1)$$

with density ρ , phase θ , and superfluid velocity proportional to $\nabla\theta$ away from vortex cores. The exact field equation is not required for the present argument. What is required is that the state of the medium has real local structure: phase, density, vortex geometry, phonon content, correlations, and boundary data.

In this paper, an “event” means a localized update of that structure. Examples include a vortex displacement, the emission or scattering of phonons, the excitation of Kelvin waves, or a reconnection. A reconnection is especially clear because it changes the connectivity of vortex filaments and converts localized order into distributed excitations. But it is not the only possible event, and it is not used here as a primitive counter for entropy.

The present state of the medium will be denoted schematically by

$$\mathcal{P}(t) = \{\rho, \theta, \mathcal{V}, \mathcal{K}, \mathcal{C}, \dots\}_t, \quad (2)$$

where $\mathcal{V}$ denotes vortex geometry, $\mathcal{K}$ Kelvin-wave content, and $\mathcal{C}$ correlation data. This notation is deliberately schematic. Its role is to keep the argument attached to present physical structure rather than to an abstract external time axis.

3 Time as Change, Not Dimension

In SSV, the physical vacuum is a real medium described by an order parameter Ψ . What physics ordinarily measures as time is not an independent dimension in which the medium is embedded, but the local rate at which the state of the medium changes.

A useful schematic statement is

$$d\tau \propto \mathcal{R}[\Psi] dt, \quad (3)$$

where $\mathcal{R}[\Psi]$ denotes the local update-rate of the medium. In later papers this quantity can be connected to gravitational and kinematic slowing. Here it is used only to express the ontological point: proper time is not a substance added to Ψ , but a measure of how much local physical change occurs in the medium.

This is not yet a final derivation, but a statement of ontology. The present is the current state of Ψ . The future does not yet exist. The past is not a region one could return to. It exists only insofar as earlier events have left structure in the present state of the medium.

3.1 Ordering Without a Block Universe

The claim that only the present state physically exists does not deny ordering. Earlier and later remain meaningful because wake structure is asymmetric. A present configuration can contain records of a previous event: outgoing phases, displaced vortex geometry, thermalized phonons, and modified reconnection opportunities. The inverse relation is not generally true. The pre-event configuration does not contain the later wake as an already-existing region.

Thus SSV keeps causal order while rejecting the reification of the past as a second place. The ordering relation is carried by structure in $\mathcal{P}(t)$, not by an external archive of completed moments.

3.2 Present-State Ontology

The strongest version of the claim is:

1. only the present state of Ψ physically exists,
2. the past is encoded in that state as record, wake, and correlation,
3. there is no second arena in which earlier states continue to exist as accessible places.

This is why time travel paradoxes are not treated here as exotic possibilities, but as signs of a mistaken description. They rely on the assumption that the past remains available as a destination.

4 Wake Production in Ψ

An event in Ψ does not simply happen and disappear. It perturbs the medium and leaves a wake. Depending on scale and context, that wake may include:

- outgoing phonon phases,
- Kelvin-wave excitations,
- vortex displacement and writhe,
- density ripples,
- altered local reconnection possibilities,
- longer-range correlations in the surrounding medium.

The important point is not that every trace remains separately visible forever. The point is that the event is redistributed into the present state of the medium rather than physically unwritten.

4.1 The Wake Is Distributed

The wake of an event is generally not stored in one location. A reconnection, for example, may leave a local change in vortex geometry, a packet of outgoing phonons, a spectrum of Kelvin waves on the retracted filaments, and altered phase correlations in the surrounding medium. After further interactions these traces may become individually hard to identify, but their information has not thereby been gathered into the unique configuration needed to restore the pre-event state.

This distributed character is what makes wake entropy different from a memory metaphor. The medium does not merely “remember” in an observer-dependent sense. Its present physical state has been changed in ways that affect what subsequent local updates are possible.

4.2 Why Reconnection Matters But Does Not Exhaust the Story

Vortex reconnection remains an important process in SSV because it is one of the clearest mechanisms by which localized structure is converted into distributed wake structure. A reconnection event changes topology, emits phonons, populates Kelvin-wave sectors, and alters future reconnection possibilities. But the present paper does not identify entropy with a bare count of reconnections. Reconnection is one important wake-writing process inside a broader picture in which any event that leaves persistent structure in Ψ contributes to the irreversible burden carried by the present state.

4.3 Local Cancellation Versus Global Unwriting

Local amplitudes can interfere destructively. Some observables can relax, thermalize, or become coarse-grained away. But this does not by itself restore the pre-event global state. To reverse an event physically would require the distributed wake to be gathered and re-focused into the unique incoming configuration that recreates the earlier state.

That is a nonlocal requirement. No local dynamical process in the medium is known to perform it.

5 Entropy as Wake-Complexity

This motivates a reframing of entropy in SSV. Entropy should not be introduced as a bare count of past events. That would risk tautology if monotonicity is built into the definition from the start. The stronger claim is:

Entropy is the irreducible wake-complexity of the present state of Ψ .

A schematic functional form is

$$S_{\text{wake}}(t) = k_B W[\Psi(t)], \quad (4)$$

where $W[\Psi(t)]$ measures the amount of present-state structure required to encode the surviving wake of prior events.

An alternative statistical form is

$$S_{\text{wake}}(t) = k_B \ln \Omega_{\text{wake}}[\Psi(t)], \quad (5)$$

where Ω_{wake} denotes the number of distinguishable wake configurations compatible with the macroscopic state.

5.1 Candidate Wake Functional

A usable functional should respond to the parts of Ψ that carry persistent wake structure. One possible schematic decomposition is

$$W[\Psi] = \alpha_\phi I_\phi + \alpha_\rho I_\rho + \alpha_\nu I_\nu + \alpha_\kappa I_\kappa + \alpha_\mathcal{C} I_\mathcal{C}, \quad (6)$$

where the I terms measure phase disorder, density disorder, vortex tangle complexity, Kelvin-wave excitation, and correlation complexity respectively. The coefficients α_i set the relative thermodynamic weight of each channel.

This is not proposed as the final entropy formula. Its purpose is to clarify what a final formula must do. It must be a functional of the present state, it must be insensitive to purely reversible relabellings, and it must increase when localized order is converted into distributed wake structure that cannot be locally unwritten.

5.2 Extensivity, Coarse-Graining, and Present Structure

The proposed wake formulation is not meant to deny the usefulness of ordinary thermodynamic entropy. Rather, it proposes a deeper physical interpretation for it. Standard thermodynamic entropy is recovered at the macroscopic level by coarse-graining over the fine wake structure of the medium. On this view, extensivity and additivity are not primitive axioms, but emergent properties of how wake structure composes across weakly coupled subsystems.

5.3 Why This Is Not Tautological

The monotonicity claim is not:

Entropy increases because we defined it to count only positive increments.

It is instead:

Events write distributed structure into Ψ , and no physical process available to the medium globally unwrites that structure back into the exact pre-event state.

This is a physical claim about the availability of inverse processes, not merely a bookkeeping choice.

5.4 Relation to Boltzmann Entropy

The standard expression $S = k_B \ln \Omega$ remains useful. The SSV proposal changes the interpretation of the microstates being counted. Instead of treating Ω as an abstract set of hidden possibilities beneath a macrostate, Ω_{wake} counts physically distinct present wake configurations compatible with the same coarse description.

This also explains why entropy can increase while the microscopic state remains definite. The medium has one actual state, but the macroscopic description loses access to the detailed distribution of wake structure. Thermodynamic entropy is then the coarse-grained shadow of a real structural burden in Ψ .

6 Formal Time Reversal and Physical Reversal

A central distinction is needed between formal and physical reversal.

6.1 Formal Reversal

In many equations, one may perform the transformation

$$t \rightarrow -t \tag{7}$$

and obtain a formally related trajectory. This is a statement about the symmetry structure of the equations.

6.2 Physical Reversal

Physical reversal would require the medium itself to perform the inverse history: gather the distributed wake, reconstruct the exact incoming phases and correlations, and restore the prior global state. The existence of a formal reverse solution does not imply the existence of a physical operation that realizes it.

This distinction is the basis for the SSV claim that time reversal may exist as a formal transformation without existing as an available process in nature.

6.3 Boundary Data and the Inverse Problem

The formal reverse of a dissipated event is not usually generated by a small local adjustment. It requires exquisitely tuned incoming boundary data across the region into which the wake has spread. In ordinary language, one would have to arrange the phonons, phases, correlations, and vortex perturbations so that they converge on the event site in precisely the configuration that reconstructs the former state.

SSV therefore locates irreversibility in the gap between a formal inverse trajectory and an available physical operation. The reverse trajectory may exist in the mathematical solution space. The present medium does not thereby contain a machine that prepares its boundary conditions.

6.4 Why This Distinction Matters

Without this distinction, the phrase “time-reversal symmetry” invites a misleading inference: that because an equation admits a formally reversed trajectory, the physical world must be able to run backward in the same sense. The SSV claim is that this inference is invalid. Formal reversal is a property of a mathematical description. Physical reversal is a claim about what operations the medium can actually perform.

7 Time-Travel Paradoxes as Category Errors

If the past exists only as structure encoded in the present state, then the usual paradoxes of time travel are misdescribed from the start. They treat the past as an existing region into which an observer may move. In SSV there is no such region. A journey to the past would mean that the current state of Ψ must somehow become both itself and an earlier state whose records it already contains.

That is not an exotic physical process. It is a contradiction in description.

8 Discussion and Open Problems

The paper makes a narrow ontological claim, but several technical tasks remain open:

1. define a computable wake functional $W[\Psi]$,

2. show explicitly how wake-complexity reduces to thermodynamic entropy under coarse-graining,
3. clarify which parts of the wake are conserved, radiated, thermalized, or erased from practical observation,
4. quantify the role of reconnection within the broader wake picture,
5. connect this time framework to gravity and black holes in later papers without overloading the present one.

The gain from the present framing is conceptual clarity. If time is local change, and the past survives only as present wake, then irreversibility is not an added rule imposed on a reversible block universe. It is the physical consequence of how events are written into a real medium.

9 Conclusion

We have argued that SSV should treat time as local change in Ψ , not as a background dimension; the past as present wake structure, not as a destination; and entropy as persistent wake-complexity, not as a bare event count. This reframing separates formal time reversal in equations from physical reversal in the medium and dissolves time-travel paradoxes as category errors. Later papers may build on this ontology, but this paper is intended to stand on its own as the cleanest statement of irreversible time in the SSV program.